ECON 5040 Midterm, In Words

For each topic: a short answer to write, using Fisher's variables and one key equation, followed by the math behind it so you actually understand it.

The exam rewards the words; the math box is there so the words make sense to you.

The recipe for every answer

(1) Name the idea. (2) Cite the one key equation and say what each side means. (3) Say what happens when something changes, and why. (4) Sketch the classic graph, if there is one.

1. Inter-temporal Prices

- *Understand how the real interest rate determines the relative price of consumption today and consumption in the future.*
- *Be able to explain the substitution incentive created by a change in the real interest rate.*

Sample response (write this)

The gross real interest rate $R = 1+r$ is the relative price of consumption today c_0 in terms of consumption tomorrow c_1: giving up one unit today buys R units tomorrow, so future consumption costs $1/R$ in today's units. The household chooses where its marginal rate of substitution between c_0 and c_1 equals R. When R rises, future consumption becomes cheaper, so the substitution effect always pushes the household to lower c_0 and save more.

Why it's true (the math)

Where the price comes from. Saving a today gives Ra tomorrow: $c_0 = e_0 - a$, $c_1 = e_1 + Ra$. Eliminating a gives $c_0 + \frac{c_1}{R} = e_0 + \frac{e_1}{R}$, so the prices are $(p_0, p_1) = (1, 1/R)$ and $R = p_0/p_1$.

FOC. Maximize $u(c_0) + \beta u(c_1)$ subject to the budget. The Lagrangian conditions are $u'(c_0) = \lambda$ and $\beta u'(c_1) = \lambda/R$; dividing gives $\frac{u'(c_0)}{\beta u'(c_1)} = R$.

Size of the substitution response (CRRA). With $u'(c) = c^{-\rho}$: $(c_1/c_0)^\rho = \beta R \Rightarrow \frac{c_1}{c_0} = (\beta R)^{1/\rho}$, so $\frac{d\log(c_1/c_0)}{d\log R} = \frac{1}{\rho}$ (the EIS). A higher ρ (more curvature) means a weaker substitution response.

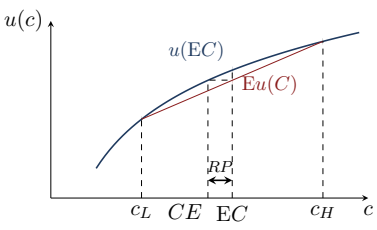
Slutsky caveat. $\partial c_0/\partial R =$ (substitution < 0) + (income effect $\propto (e_0 - c_0)$). For a lender ($e_0 > c_0$) the income effect is positive, so the *total* effect on saving is ambiguous; only the substitution effect is signed.

2. Choice Under Uncertainty

- *Understand expected utility, risk aversion, and how the curvature of the utility function affects the value of a risky prospect.*
- *Be able to compute and interpret the certainty equivalent of a simple gamble.*

Sample response (write this)

Under expected utility, a gamble is valued by the probability-weighted average of the utility u of each outcome, and a concave u means risk aversion: the sure average is preferred to the gamble. The certainty equivalent CE is the sure amount that gives the same expected utility; for the 50/150 gamble with log utility, $CE \approx 86.60$ even though the mean is 100, so the risk premium $RP \approx 13.40$. The more curved u is, the lower the CE and the larger the RP.



Why it's true (the math)

CE computation. $\log CE = \frac{1}{2}\log 50 + \frac{1}{2}\log 150 = \log\sqrt{7500}$, so $CE = 86.60$. In general $CE = u^{-1}(\mathrm{E}u(C))$; for log utility it is the geometric mean, and for CRRA it is $CE = \big(\mathrm{E}[C^{1-\rho}]\big)^{1/(1-\rho)}$.
Jensen. If $u'' < 0$ then $\mathrm{E}u(C) < u(\mathrm{E}C)$, so $u(CE) < u(\mathrm{E}C)$ and, because u is increasing, $CE < \mathrm{E}C$.
Arrow–Pratt (why curvature sets the premium). For a small risk $C = \bar{c} + \varepsilon$ with $\mathrm{E}\varepsilon = 0$ and $\mathrm{Var}\varepsilon = \sigma^2$, Taylor-expand both sides of $u(\bar{c} - RP) = \mathrm{E}u(\bar{c} + \varepsilon)$: $u - u'RP \approx u + \frac{1}{2}u''\sigma^2$, so

$$RP \approx \tfrac{1}{2}\Big(-\frac{u''(\bar{c})}{u'(\bar{c})}\Big)\sigma^2 = \tfrac{1}{2}A(\bar{c})\,\sigma^2, \qquad \rho(c) = cA(c).$$

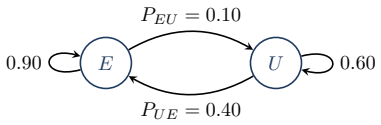
For log, $A = 1/c$ and $\rho = 1$; for CRRA, ρ is constant. Check: $\frac{1}{2}\cdot\frac{1}{100}\cdot 50^2 = 12.5 \approx 13.4$.
vNM uniqueness. Only positive affine transforms $a + bu$ preserve the ranking, since $\mathrm{E}[a + bu] = a + b\mathrm{E}u$.

3. Markov Processes

- *Understand the meaning of the Markov property: conditional on the current state, the past provides no additional information about the distribution of the next state.*
- *Know how to interpret a transition matrix P, including the economic meaning of an element P_{ij} and of an entire row.*

Sample response (write this)

A process is Markov if, once you know today's state X_t, the earlier history adds nothing to the forecast of tomorrow's state X_{t+1}. The element P_{ij} is the probability of moving to state j tomorrow given state i today (for example, the unemployed-to-employed entry is the job-finding rate), and row i is the entire distribution of tomorrow's state starting from i, so its entries sum to one.



Why it's true (the math)

Distributions. $\psi_{t+1}(j) = \sum_i \psi_t(i)P_{ij}$, i.e. $\psi_{t+1} = \psi_t P$ and $\psi_t = \psi_0 P^t$.
Multi-step (Chapman–Kolmogorov). $(P^2)_{ij} = \sum_k P_{ik}P_{kj}$, summing over intermediate states. With $P = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix}$: $(P^2)_{EU} = .9(.1) + .1(.6) = .15$.
Stationary distribution. $\psi^* = \psi^* P$. Unemployment flows balance: inflow $e \cdot P_{EU}$ equals outflow $u \cdot P_{UE}$, so $.1e = .4u$; with $e + u = 1$ this gives $u^* = \frac{P_{EU}}{P_{EU}+P_{UE}} = \frac{.1}{.5} = 0.2$.
Why rows sum to 1. $\sum_j P_{ij} = \Pr(X_{t+1} \in \text{anything} \mid X_t = i) = 1$. Columns don't have to.

4. Introduction to Dynamic Programming

- *Understand the Bellman equation as a decomposition of a dynamic problem into a current reward and a discounted expected continuation value.*
- *Be able to identify the state variable, choice variable, current payoff, and next-period state in a dynamic economic problem.*

Sample response (write this)

The Bellman equation says the value $V(x)$ of being in state x equals the best choice u of today's reward $r(x,u)$ plus the discounted (β) expected value of tomorrow's state x', assuming optimal behavior from then on. In the growth model, the state is current capital k, the choice is next period's capital k', the current payoff is the utility of consuming whatever output is not saved, and the next-period state is k'.

Why it's true (the math)

Why it works (principle of optimality). Separate out the $t=0$ term of the sequence problem:

$$\max_{\{u_t\}} \mathrm{E}\sum_{t\geq 0}\beta^t r(x_t,u_t) = \max_{u_0}\Big\{r(x_0,u_0) + \beta\,\mathrm{E}\Big[\underbrace{\max_{\{u_t\}_{t\geq 1}} \mathrm{E}\sum_{t\geq 1}\beta^{t-1}r(x_t,u_t)}_{=V(x_1)}\Big]\Big\}.$$

The tail problem starting at x_1 has the same form, so its value is $V(x_1)$; that's the recursion.
Guess and verify (growth). Guess $V = E + F\log k$. The FOC $\frac{1}{Ak^\alpha - k'} = \frac{\beta F}{k'}$ gives $k' = \frac{\beta F}{1+\beta F}Ak^\alpha$. Matching the $\log k$ coefficients gives $F = \alpha + \alpha\beta F \Rightarrow F = \frac{\alpha}{1-\alpha\beta}$, so $k' = \alpha\beta Ak^\alpha$ and $k^* = (\alpha\beta A)^{1/(1-\alpha)}$.
Euler from Bellman. Combining the FOC $u'(c_t) = \beta V'(k_{t+1})$ with the envelope condition $V'(k) = u'(c)f'(k)$ gives $u'(c_t) = \beta u'(c_{t+1})f'(k_{t+1})$.

5. Value Functions and Policy Functions

- *Understand the distinction between a value function and a policy function.*
- *Know what economic information each function provides and how the optimal policy is related to the Bellman equation.*

Sample response (write this)

The value function $V(x)$ tells you how good it is to start in state x and act optimally from then on, which is a measure of welfare. The policy function $h(x)$ tells you which action to take in each state, a complete contingent plan. The optimal policy is the choice that maximizes the right-hand side of the Bellman equation, and following h forever delivers exactly V.

Why it's true (the math)

From V to h: a single maximization in each state. **From h to V_h:** with the policy fixed there's no max, so $V_h = r_h + \beta P_h V_h \Rightarrow V_h = (I-\beta P_h)^{-1}r_h = \sum_{t\geq 0}\beta^t P_h^t r_h$ (a Neumann series: the expected discounted reward stream).
Economic content of V. By the envelope condition $V'(x) = u'(c)$ (cake) or $u'(c)f'(k)$ (growth), so V' is the shadow price of one more unit of the state.
Economic content of h. The law of motion $x_{t+1} \sim g(x_t, h(x_t))$; in a finite DP it induces the Markov chain $P_h(x,\cdot) = M_{h(x)}(x,\cdot)$, whose ψ^* gives long-run outcomes. Examples: $h(k) = \alpha\beta Ak^\alpha$; $c^*(x) = (1-\beta^{1/\rho})x$; McCall's "accept iff $w \geq \bar{w}$."

6. Howard's Method versus Value Function Iteration

- *Understand the basic logic of value function iteration: repeatedly applying the Bellman operator until the value function converges.*
- *Understand how Howard's method separates policy evaluation from policy improvement, and why this can make computation much faster.*

Sample response (write this)

Value function iteration starts from a guessed value function V_0 and repeatedly applies the Bellman operator T, adding one more period of future rewards each round, until V stops changing; the policy h is read off at the end. Howard's method starts from a guessed policy h, evaluates it exactly (the value of following h forever, one linear solve), then improves it by picking the best action in each state, and stops when h no longer changes. Howard is faster because each evaluation does the work of infinitely many VFI steps, so it finishes in a few rounds.

Why it's true (the math)

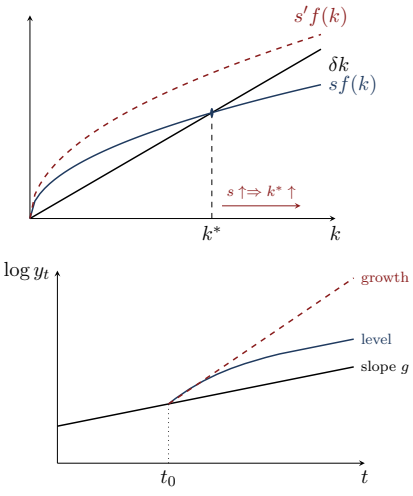
VFI convergence. T is a contraction: $\|TV - TW\|_\infty \leq \beta\|V - W\|_\infty$ (Blackwell: monotonicity plus discounting). So $\|V_n - V^*\| \leq \beta^n\|V_0 - V^*\|$. To cut the error by 10^{-6} you need $n \approx \frac{\ln 10^{-6}}{\ln \beta}$: about 270 steps at $\beta = .95$ and about 1,400 at $\beta = .99$.
Howard convergence. Policy improvement is monotone, $V_{h_{k+1}} \geq V_{h_k}$, and there are at most $|U|^{|X|}$ policies ($2^3 = 8$ for the student problem), so it terminates in finitely many steps, usually just a handful. The trade-off is that each step solves an $|X| \times |X|$ linear system.
Student check ($\beta = .5$, $h_0 =$ TV): $V_{h_0} \approx (3.64, 6.92, 13.40)$. Sick: TV $1 + .5(.5 \cdot 3.64 + .5 \cdot 6.92) = 3.64$ vs. Study $-1 + .5(.9 \cdot 3.64 + .1 \cdot 6.92) = 0.98$, so keep TV; the same holds in every state, so it stops at once.

7. The Solow Model

- *Understand how saving and capital accumulation determine the steady-state levels of capital and output.*
- *Be able to distinguish a change in the level of steady-state output from a permanent change in the long-run growth rate.*

Sample response (write this)

In Solow, a fixed fraction s of output is invested and a fraction δ of capital depreciates. Because of diminishing returns, investment eventually just replaces depreciation, so capital per worker converges to a steady state k^* with output y^*. A higher s raises k^* and y^* (a **level effect**), but long-run growth of output per worker equals the rate of technological progress g, so saving does not change it.



Why it's true (the math)

Steady state. $sAk^\alpha = \delta k \Rightarrow k^{1-\alpha} = sA/\delta$. Then $\log y^* = \log A + \frac{\alpha}{1-\alpha}\log\frac{sA}{\delta}$, so $\frac{\partial \log y^*}{\partial \log s} = \frac{\alpha}{1-\alpha}$ (about 0.5 with $\alpha = 1/3$). This elasticity is a *level* shift.
Why growth dies out. $\frac{k_{t+1}-k_t}{k_t} = sAk_t^{\alpha-1} - \delta \to 0$ as $k_t \to k^*$, since $k^{\alpha-1} \downarrow$ (diminishing MPK). Any one-time increase in s is eventually absorbed.
Balanced growth. With $E_{t+1} = (1+g)E_t$, $\tilde{k} = K/(EL) \to \tilde{k}^*$, so $Y/L = E\tilde{y}^*$ grows at g and Y grows at $n + g + ng$; s shows up only in $\tilde{k}^*$.
Speed. $\lambda = 1 - \frac{(1-\alpha)(\delta+n+g+ng)}{(1+n)(1+g)} \approx .948$ and half-life $= \frac{\ln .5}{\ln \lambda} \approx 13$ years, independent of s.
Golden rule. $\max_s (1-s)f(k^*)$ gives $f'(k^*) = \delta$, so $s_{GR} = \alpha$.

8. Optimal Saving

- *Understand the Euler equation $u'(c_t) = \beta R \mathrm{E}_t[u'(c_{t+1})]$ as a comparison between the marginal value of consuming today and the expected discounted marginal value of saving for tomorrow.*
- *Be able to explain how a household changes consumption and saving when the two sides of the Euler equation are not equal.*

Sample response (write this)

The left side is the marginal utility of consuming one more unit today, c_t; the right side is the value of saving it instead: it grows to R units, is valued at tomorrow's expected marginal utility of c_{t+1}, and is discounted by β. If the left side is bigger, the household consumes more today and saves less; if it is smaller, it saves more. Either adjustment continues until the two sides are equal, unless a borrowing constraint binds.

Why it's true (the math)

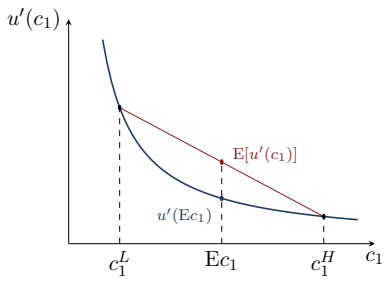
Perturbation argument. Cut c_t by ε and save it, then raise c_{t+1} by $R\varepsilon$. The utility change is $\Delta \approx \varepsilon\left[-u'(c_t) + \beta R \mathrm{E}_t u'(c_{t+1})\right]$. At an optimum no ε can help, so the bracket is 0. If the bracket is < 0 (LHS $>$ RHS), $\varepsilon < 0$ helps: consume more now.
Why it converges. Raising c_t lowers $u'(c_t)$, and because c_{t+1} falls, $u'(c_{t+1})$ rises ($u'' < 0$). The gap closes.
From Bellman (cake, $R = 1$). The FOC $u'(c) = \beta V'(x-c)$ plus the envelope condition $V'(x) = u'(c)$ give $u'(c_t) = \beta u'(c_{t+1})$. Under CRRA, $c_{t+1}/c_t = (\beta R)^{1/\rho}$; cake eating gives $c^*(x) = (1-\beta^{1/\rho})x$ (sum the geometric series $x = c\sum_j q^j$ with $q = \beta^{1/\rho}$).
Borrowing limit. The KKT conditions are $u'(c_t) = \beta R \mathrm{E}u'(c_{t+1}) + \lambda$ with $\lambda \geq 0$ and $\lambda(a' - \underline{a}) = 0$, so $u'(c_t) > \beta R \mathrm{E}u'(c_{t+1})$ when the constraint binds.

9. Income Fluctuations

- *Understand precautionary saving: uncertainty about future income can increase desired saving even when expected future income is unchanged.*
- *Understand why assets provide self-insurance by transferring resources toward future states in which consumption and wealth are low.*

Sample response (write this)

When future income Y_1 is risky, low-income states have very high marginal utility, so a unit saved is especially valuable as protection, and the household saves more than it would with the same average income for sure. This is **precautionary saving**. Assets a pay the same return R in every state, but since consumption is chosen after income is known, the household draws down its buffer when income is low, moving resources to the states where consumption and wealth are low (**self-insurance**).



Why it's true (the math)

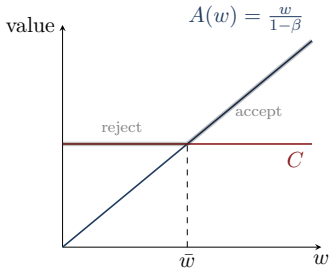
Jensen on u'. For log utility $u'(c) = 1/c$, which is convex, so $\mathrm{E}\left[\frac{1}{Ra_1+Y_1}\right] > \frac{1}{Ra_1+\mathrm{E}Y_1}$. At any fixed a_1, risk raises the RHS of the Euler equation, so the optimal a_1 must rise to restore equality (raising $u'(c_0)$ and lowering c_1's marginal utility).
Prudence (second order). $\mathrm{E}u'(\bar{c}+\varepsilon) \approx u'(\bar{c}) + \frac{1}{2}u'''(\bar{c})\sigma^2$. The extra term is positive iff $u''' > 0$ (CRRA: $u''' = \rho(\rho+1)c^{-\rho-2} > 0$). Quadratic utility has $u''' = 0$, which means no precautionary saving: *concavity alone isn't enough.*
Example. $y_0 = 10$, $\beta = .95$, $R = 1.02$, $Y_1 \in \{2, 10\}$ vs. a certain 6: the risky case gives a larger a_1.
Infinite horizon. $V(a, z) = \max_{a' \geq \underline{a}}\{u(Ra + y(z) - a') + \beta \sum_{z'} P(z, z')V(a', z')\}$. The state is (a, z) because z forecasts y' when income is persistent; it combines DP, a Markov chain, and the Euler equation.

10. Putting the Ideas Together: Job Search

- *Understand why job search is a dynamic decision problem: today's decision affects both today's payoff and the opportunities available tomorrow.*
- *Be able to formulate a job-search problem in terms of state variables, a choice variable, the transition to tomorrow's state, and a continuation value.*

Sample response (write this)

Job search is dynamic because rejecting an offer gives up today's wage for unemployment compensation c but keeps the option of a new random offer tomorrow. The state is the current offer w, the choice is accept or reject, and the transition is: accept means earning w forever, reject means drawing a new offer next period. The continuation value C is the value of rejecting and searching optimally again; since the value of accepting rises with w while C doesn't depend on w, the worker accepts any offer above the reservation wage $\bar{w}$.



Why it's true (the math)

Accept value. $A(w) = \sum_{t \geq 0} \beta^t w = \frac{w}{1-\beta}$.
Reservation-wage equation. At $\bar{w}$, $\frac{\bar{w}}{1-\beta} = C$, and $V(w') = \max\{\frac{w'}{1-\beta}, \frac{\bar{w}}{1-\beta}\}$. Substituting into C:

$$\frac{\bar{w}}{1-\beta} = c + \frac{\beta}{1-\beta}\mathrm{E}\max\{W', \bar{w}\} \quad \Longrightarrow \quad \bar{w} - c = \frac{\beta}{1-\beta}\mathrm{E}\left[(W' - \bar{w})^+\right].$$

The LHS is the cost of searching one more period (forgoing $\bar{w}$ for c); the RHS is the expected gain from a better draw, valued forever.
Comparative statics from that equation. $c \uparrow \Rightarrow \bar{w} \uparrow$. $\beta \uparrow$ raises the RHS $\Rightarrow \bar{w} \uparrow$. A mean-preserving spread raises $\mathrm{E}(W' - \bar{w})^+$ (a convex function of W') $\Rightarrow \bar{w} \uparrow$: option value.

Computing C. Iterate $C_{n+1} = c + \beta \mathrm{E} \max\{\frac{W'}{1-\beta}, C_n\}$; this is VFI on a scalar.